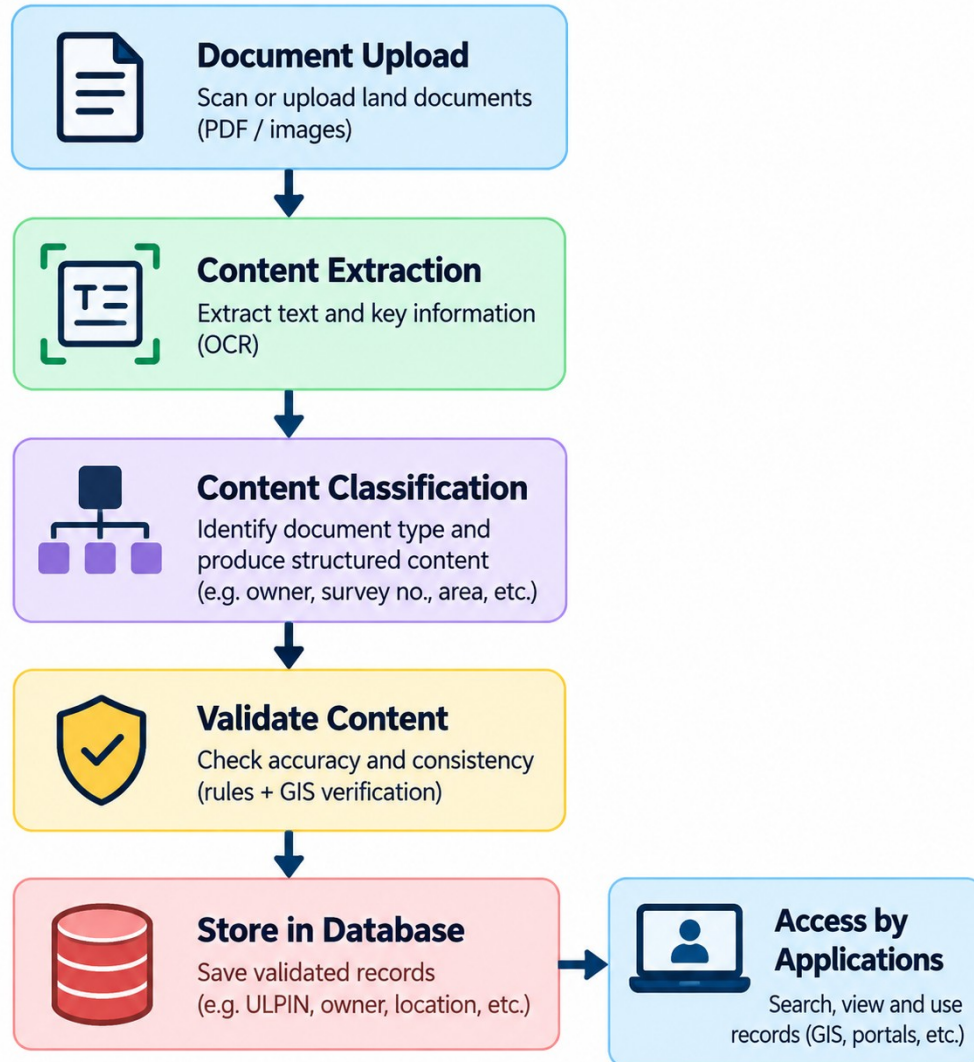


TITLE PAGE

- Problem Statement ID- **SIH26018**
- Problem Statement Title- *Intelligent Land Record Digitization and Validation System*
- Theme- **Smart Automation**
- PS Category- **Software**
- Team ID-
- Team Name- **PrithviX**



Intelligent Land Record Digitization and Validation System



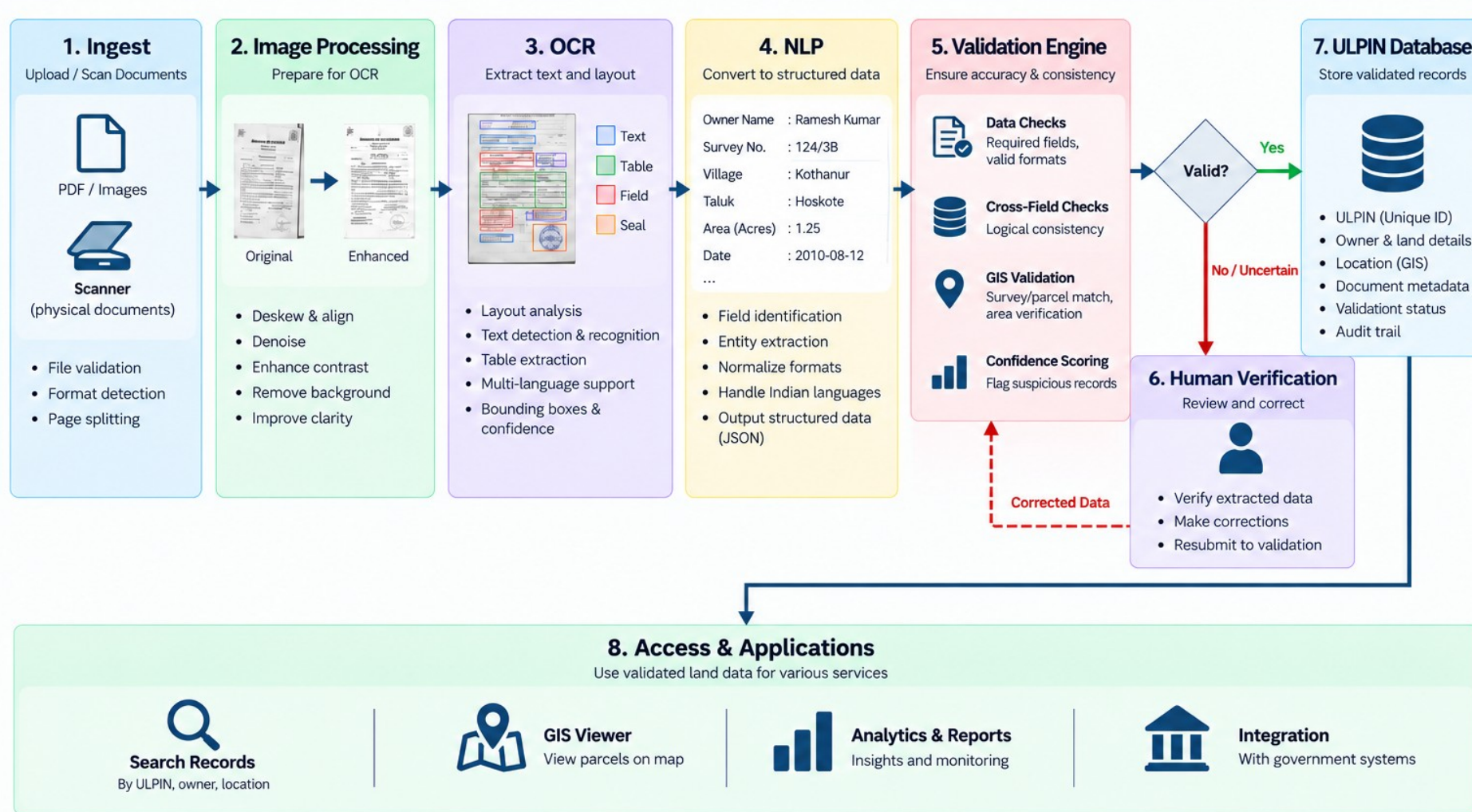
How it addresses the problem?

Digitizes and stores legacy land documents by **centralising them into one unified platform**, making them easily searchable and accessible.



What's innovative?

Uses **AI-based data classification and validation** to automatically understand and verify document information, improving accuracy and consistency.



Tech Stack



Frontend

React
(JavaScript)



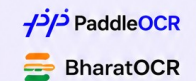
Backend

Python, Flask



Image Processing

OpenCV



OCR

PaddleOCR /
BharatOCR



AI/NLP

Data Classification
& Extraction



Validation/GIS

Rule Engine,
PostGIS



Database

PostgreSQL +
PostGIS

Analysis of Feasibility

The solution is technically feasible using established technologies such as OCR, AI/NLP, image processing, GIS, and centralized databases. Its modular pipeline allows document processing, extraction, classification, validation, and storage to be developed and tested independently.

Potential Challenges & Risks

Legacy documents may be poor-quality, damaged, handwritten, multilingual, or inconsistently formatted, which can reduce extraction accuracy. Conflicting land information may also make automated validation difficult, while large-scale digitization can increase processing and human-review requirements.

Strategies for Overcoming Challenges

Image pre processing can improve OCR quality, while AI-based classification combined with rule-based and GIS validation can detect inconsistencies. Low-confidence records can be routed for human verification, and the modular architecture allows individual components to be improved or scaled without redesigning the entire system.

Potential Impact on Target Audience

Making land information accessible, reliable and useful for everyone



Citizens / Landowners

- Easier access to land records
- Reduced need for physical visits
- Greater transparency and trust

"Quick and hassle-free access to my land records"



Government Departments

- Digitized and standardized records
- Faster verification and decision making
- Reduced manual workload

"Reliable data for better governance"



Surveyors / Field Officials

- Easy access to GIS-linked records
- Accurate field verification
- Streamlined reporting

"Verified information at my fingertips"



Financial Institutions

- Trusted and verifiable land data
- Faster loan processing
- Reduced risk of fraud

"Reliable data for safer lending"



Society at Large

- Transparent land governance
- Reduced disputes and corruption
- Inclusive access to land information

"A fairer and more transparent land ecosystem"

Benefits of the Solution

Digitized land records for a more transparent, efficient and sustainable future



Social Benefits

Empowering people, building trust

- Greater transparency in land transactions
- Reduced land disputes
- Equal access to information for all citizens
- Stronger citizen-government trust



Economic Benefits

Higher efficiency, greater opportunities

- Faster processing of loans and approvals
- Reduced fraud and illegal transactions
- Lower administrative costs
- Improved land use planning and investment opportunities



Environmental Benefits

Smarter land use, a more sustainable tomorrow

- Better land use planning with GIS data
- Protection of ecological and sensitive areas
- Reduced encroachment on forests and public land
- Support for sustainable development



Governance Benefits

Data-driven, accountable and efficient

- Centralized and standardized land records
- Faster verification and decision making
- Improved inter-departmental coordination
- Audit trails and better accountability

Accurate Records | Transparent Governance | Stronger Communities

- **Digital India Land Records Modernization Programme (DILRMP)**
dolr.gov.in/en/document-category/program-dilrmp
- **DILRMP MIS 4.0**
dilrmp.gov.in
- **DILRMP 3.0 Operational Guidelines (2026–2031)**
cdnbbsr.s3waas.gov.in/s3d69116f8b0140cdeb1f99a4d5096ffe4/uploads/2026/08/20260827600317271.pdf
- **ULPIN / Bhu-Aadhaar Press release**
www.pib.gov.in/Pressreleaseshare.aspx?PRID=1932867&lang=2®=48
- **Paddle OCR**
www.paddleocr.ai/main/en/index/index.html
- **Open CV**
github.com/opencv/opencv